

Explanation vs prediction; reporting

Workflow lab: Goal → Approach → Execution → Check → Report.

Learning objectives

- Distinguish explanatory from predictive modelling, and choose the correct evaluation metric for each.
- Map STROBE, TRIPOD, STARD, and CONSORT onto the study designs each is meant for.
- Produce a publication-ready regression table with `gtsummary` whose numbers trace directly to the fitted model.

Prerequisites

The regression tools covered earlier in the book.

Background

Breiman's 1998 essay *The Two Cultures* and Shmueli's 2010 essay *To Explain or to Predict?* draw the same line from opposite sides. In explanatory modelling the goal is inference — what is the estimated effect of X on Y, adjusted for confounders? — and the right evaluation metric is bias-unbiasedness, interval coverage, and interpretability. In predictive modelling the goal is accuracy on unseen data, and the right evaluation metric is out-of-sample loss, calibration, and decision utility. The statistical machinery overlaps but the habits around it do not; a regression that makes an excellent explanation often makes a lacklustre prediction, and vice versa.

Reporting guidelines are the mechanism by which the field enforces discipline around these two cultures. STROBE for observational studies, CONSORT for randomised trials, STARD for diagnostic accuracy, and TRIPOD (now TRIPOD-AI) for prediction models each specify the items a reader needs to evaluate the claim. A checklist filled in as you write is much easier than one filled in at submission.

Setup

```
library(tidyverse)
library(gtsummary)
library(broom)
library(palmerpenguins)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Goal

Fit a single linear model to the penguins data and present it two ways: once as an explanatory analysis (what drives body mass?) and once as a predictive model (can we predict body mass on held-out birds?).

2. Approach

```
fit_expl <- lm(body_mass_g ~ flipper_length_mm + sex + species, data = peng)
```

3. Execution

```
peng |>
  ggplot(aes(flipper_length_mm, body_mass_g, colour = species)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, linewidth = 0.8) +
  labs(x = "Flipper length (mm)", y = "Body mass (g)")
```

4. Check

Explanatory evaluation: effect sizes with intervals, residual diagnostics, and adjusted R^2 .

Predictive evaluation: a 5-fold split, hand-coded to keep the example transparent.

```
folds <- sample(rep(seq_len(k), length.out = nrow(peng)))
rmse_k <- sapply(seq_len(k), function(i) {
  tr <- peng[folds != i, ]
  te <- peng[folds == i, ]
  f <- lm(body_mass_g ~ flipper_length_mm + sex + species, data = tr)
  sqrt(mean((predict(f, te) - te$body_mass_g)^2))
})
mean_rmse <- mean(rmse_k)
mean_rmse
```

5. Report

```
tbl_regression(fit_expl, intercept = TRUE) |>
  modify_caption("**Table 1. Linear-regression estimates for body mass (g).**")
```

In the Palmer penguins dataset ($n = \text{nrow}(\text{peng})$), body mass was associated with flipper length, sex, and species (adjusted $R^2 = \text{round}(\text{summary}(\text{fit_expl})\$adj.r.squared, 2)$). Out-of-sample performance, estimated by 5-fold cross-validation, was $\text{RMSE} = \text{round}(\text{mean_rmse}, 0)$ g.

Reporting-guideline map

Design	Guideline	URL
Randomised trial	CONSORT	https://www.consort-statement.org/
Observational study	STROBE	https://www.strobe-statement.org/
Diagnostic-accuracy study	STARD	https://www.equator-network.org/reporting-guidelines/stard/
Prediction-model study	TRIPOD / TRIPOD-AI	https://www.tripod-statement.org/
Systematic review	PRISMA	http://prisma-statement.org/

Common pitfalls

- Choosing a predictor set by R^2 and reporting a predictive claim, or tuning a predictive model and reporting causal-sounding coefficients.
- Filling in a reporting checklist at submission rather than while drafting.
- Assuming in-sample R^2 generalises.

Further reading

- Breiman L. (2001). *Statistical Modeling: The Two Cultures*.
- Shmueli G. (2010). *To Explain or to Predict?* Statistical Science.
- Writing a report.

Session info
